RAJALAKSHMI ENGINEERING COLLEGE

Praveen A 250701547-2025-CSE

P2

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|---|---|
| **Started on** | Sunday, 30 August 2026, 12:47 PM |
| **State** | Finished |
| **Completed on** | Sunday, 30 August 2026, 12:51 PM |
| **Time taken** | 4 mins 2 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Given two arrays array_One[] and array_Two[] of same size N. We need to first rearrange the arrays such that the sum of the product of pairs( 1 element from each) is minimum. That is SUM (A[i] * B[i]) for all i is minimum.

**For example:**

| Input | Result |
|---|---|
| 3<br>1<br>2<br>3<br>4<br>5<br>6 | 28 |

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int compareAsc(const void *a, const void *b) {
    int valA = *(const int *)a;
    int valB = *(const int *)b;
    if (valA > valB) return 1;
    if (valA < valB) return -1;
    return 0;
}

int compareDesc(const void *a, const void *b) {
    int valA = *(const int *)a;
    int valB = *(const int *)b;
    if (valB > valA) return 1;
    if (valB < valA) return -1;
    return 0;
}

int main() {
    int n;
    if (scanf("%d", &n) != 1 || n <= 0) return 0;

    int *array_One = (int *)malloc(n * sizeof(int));
    int *array_Two = (int *)malloc(n * sizeof(int));

    for (int i = 0; i < n; i++) {
        scanf("%d", &array_One[i]);
    }
    for (int i = 0; i < n; i++) {
        scanf("%d", &array_Two[i]);
    }

    qsort(array_One, n, sizeof(int), compareAsc);
    qsort(array_Two, n, sizeof(int), compareDesc);

    long long min_product_sum = 0;
    for (int i = 0; i < n; i++) {
        min_product_sum += (long long)array_One[i] * array_Two[i];
    }

    printf("%lld\n", min_product_sum);

    free(array_One);
    free(array_Two);
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 3<br>1<br>2<br>3<br>4<br>5<br>6 | 28 | 28 | ✔ |
| ✔ | 4<br>7<br>5<br>1<br>2<br>1<br>3<br>4<br>1 | 22 | 22 | ✔ |
| ✔ | 5<br>20<br>10<br>30<br>10<br>40<br>8<br>9<br>4<br>3<br>10 | 590 | 590 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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